NORMATIVE SYSTEMS ARCHITECTURE & ENGINEERING SPECIFICATION · SPEC-WEBIZEN-VPS-0.0.40

Webizen & QualiaDB Sovereign VPS Hosting Platform

Comprehensive Requirements, Architecture & Implementation Plan for Large Ontology Processing (>100 MB **`WebCivics/ns`**), Multi-Tenant Sandboxing, QDNF Naming/DNS Acceleration, User-Powered Decentralized CDN, and Permissive Commons Micropayments

Specification Attribute	Normative Baseline & Repository Alignment
Target System & Scope	Turnkey Linux VPS Hosting Stack (cPanel/WHM replacement) & Online Extension Relay for local webizen-desktop / QualiaDB installations (`:8080` ↔ `:4242`)
Primary Seed Workload	Bounded streaming ingestion, `.q42` v4 / `QBDL .hmc` compilation, and cost-capped serving of https://github.com/WebCivics/ns (>100 MB) and ns.webcivics.net human-rights instruments
Codebase Baseline	QualiaDB 0.0.40-dev (`qualia-core-db`, `qualia-client-core`, `vibe`, `qualia-peer`, `qualia-solid-bridge`, `qualia-cooperative-core`, `poet`)
Stewardship & Governance	Single Natural Person, Family/Household (`GuardianshipSemantics`), or Multi-Person Community/Cooperative (`M-of-N` Steward Quorum); strict separation of **Who-Plane (`NaturalAgent`)** from **Handle-Plane (DNS/ABN)**
Economic & CDN Model	Permissive Commons (`modalities/value_flow.rs`): $0 for Natural Persons & Humanitarian ICT; Commercial Tolls up to `10×` Obligation Cost with automatic Commons Sunset + User-Powered Peer CDN with ILP-STREAM & eCash/DLT Micropayments

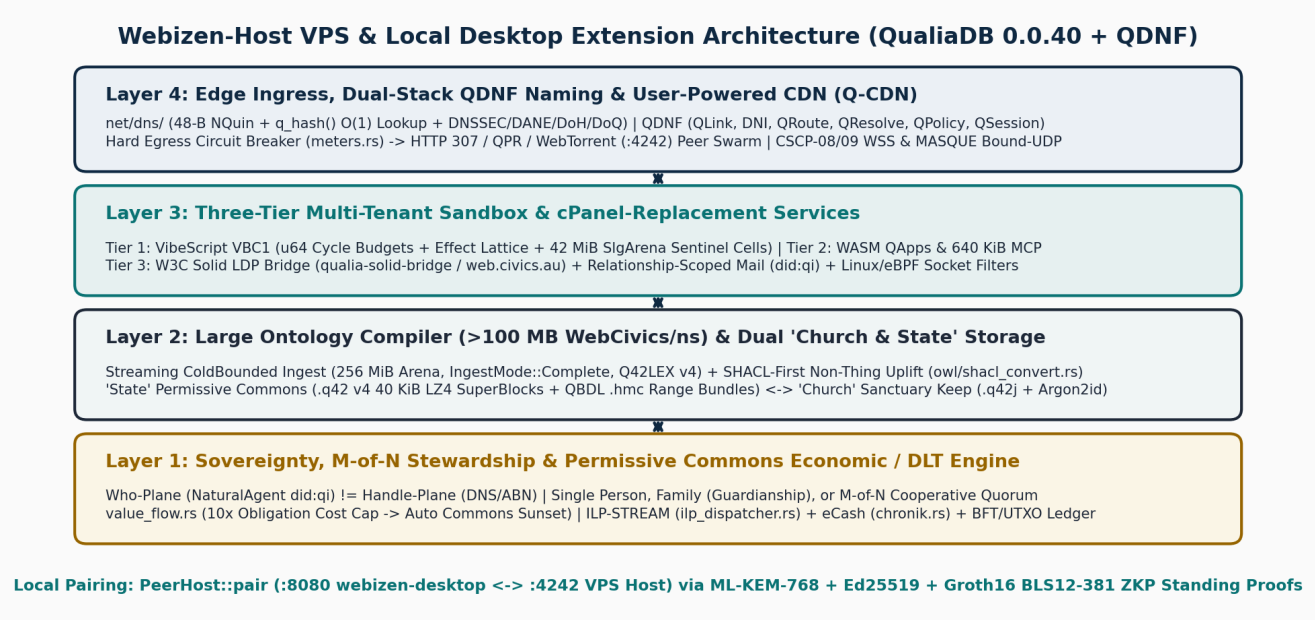


Figure 1: Four-layer architecture of the Webizen-Host VPS stack, illustrating how large ontology compilation (`.q42` v4 / `QBDL .hmc`), Three-Tier tenant sandboxing, QDNF/`NQuin` DNS resolution, and the User-Powered Decentralized CDN integrate directly with `qualia-core-db` and paired `webizen-desktop` clients.

Prepared for: Timothy Charles Holborn (Webizen / QualiaDB & WebCivics) | **Date:** October 26, 2026 | **Authority Contract:** Grounded against live `qualia-core-db 0.0.40` crate layouts, QDNF 0.0.39 evidence rules, and Project Definitions.

1. Executive Summary: Sovereign Hosting Beyond Cloud Rent-Seeking

Operating foundational ontological infrastructure and human-centric web services on conventional cloud platforms exposes independent stewards to unbounded financial liability and architectural lock-in. The operational history of ns.webcivics.net—which provides machine-readable human rights instruments coupled with formal deontic and defeasible logic—demonstrated two structural failure modes in legacy cloud hosting. First, standard web repositories and semantic tools fail when importing large ontological source repositories exceeding 100 MB such as https://github.com/WebCivics/ns, forcing clients to download monolithic text files that cannot be parsed on mobile or browser devices. Second, commercial content delivery networks such as AWS CloudFront provide no protocol-level cost circuit breaker when high-volume automated traffic arrives from major US data hubs, turning public-good knowledge stewardship into an uncapped financial liability.

The **Webizen Web-Host Server Stack (`webizen-host`)** is a comprehensive Virtual Private Server (VPS) hosting platform and full cPanel/WHM replacement engineered natively on the QualiaDB and Webizen stack, as defined in the Webhost Project Definitions. Designed for turnkey installation on commodity Linux VPS instances or as an online federated extension to a locally installed webizen-desktop and QualiaDB environment, the platform unifies five critical capabilities within a single Rust workspace: (1) bounded streaming compilation and range-addressable hosting of massive ontologies in `.q42` v4 and `QBDL .hmc` formats; (2) individual, family, and multi-person cooperative server ownership with strict *Church & State* vault separation; (3) three-tier application sandboxing powered by VibeScript (`VBC1`), WebAssembly, and W3C Solid LDP compatibility (web.civics.au); (4) sub-microsecond DNS and naming resolution via `net/dns/` 48-byte `NQuin` records and the Qualia Decentralized Network Fabric (QDNF); and (5) a user-powered decentralized CDN with hard origin cost caps and protocol-level micropayments.

Platform Dimension	Legacy Stack Failure Mode (cPanel / CloudFront / RDF)	Webizen-Host Specification (QualiaDB 0.0.40 Native)	Authoritative Repo Locus
Large Ontology Processing (>100 MB)	Monolithic RDF/XML & Turtle files exceed 100 MB import limits and crash browser/mobile memory.	Streaming `ColdBounded` compiler produces `.q42` v4 volumes (40 KiB LZ4 SuperBlocks of 48-B `NQuins` + `Q42LEX v4` paged lexicon) & `QBDL .hmc` shards fetched via HTTP/QPR byte-range slices.	`qualia-core-db/src/q42/`, `q42_lex_ns.rs`, `q42/asset_import/`, `bundle/format.rs`
Ownership & Legal Continuity	Single corporate root admin; losing an ICANN domain erases user identity and service access.	Owned by a Natural Person, Family (`GuardianshipSemantics`), or Multi-Person Group (`M-of-N` Steward Quorum). **Who-Plane (`NaturalAgent`)** is never tied to **Handle-Plane (DNS/ABN)**.	`qualia-cooperative-core`, `stewardship.rs`, `foundation/crdt.rs`, `did_qi/`
Service & Website Sandboxing	Heavy VMs or shared PHP/Node.js processes vulnerable to cross-tenant memory/CPU starvation.	**3-Tier Sandbox:** (1) VibeScript `VBC1` with `u64` cycle budgets & 42 MiB `SlgArena` cells, (2) WASM QApps, (3) Linux cgroups/namespaces + eBPF socket filters + Solid LDP bridge.	`crates/vibe/`, `governance/cell_graph.rs`, `qualia-solid-bridge`, `net/` eBPF
DNS & Naming Resolution	Heap-heavy text zone parsing; DNS treated as identity authority; slow recursive lookups.	48-byte `NQuin` DNS zone ring with compile-time `q_hash()` FNV-1a O(1) lookup, deontic/SHACL policy filtering, DNSSEC/DANE/DoH/DoQ, and native `QDNF` (`QResolve` / `DNI`).	`net/dns/`, `net/qdnf/resolve/`, `governance/web_civics.rs`
CDN & Egress Cost Protection	CloudFront/commercial CDNs charge uncapped per-GB egress; US hub scraping causes bill shock.	Hard egress byte/dollar circuit breakers (`meters.rs`) automatically offload bulk `.q42`/`.hmc` shards to the **User-Powered Peer CDN** (`QPR` + `:4242` WebTorrent seeder + OPFS cache).	`net/qdnf/economics/meters.rs`, `net/peer/replication/`, `services/webtorrent_*.rs`

Platform Dimension	Legacy Stack Failure Mode (cPanel / CloudFront / RDF)	Webizen-Host Specification (QualiaDB 0.0.40 Native)	Authoritative Repo Locus
Commons Economics & DLT	Unfunded open-source charity or perpetual SaaS rent extraction.	**Permissive Commons:** $0 for Natural Persons / Humanitarian ICT; Commercial Tolls recover up to `10×` Obligation Cost then auto-sunset; peer CDN micropayments via ILP-STREAM & eCash/DLT.	`modalities/value_flow.rs`, `services/ilp_dispatcher.rs`, `chronik.rs`

2. Deployment Topologies & Stack-Improvement Symbiosis

As specified in the Project Definitions and the Webizen / QualiaDB Reorganisation Notes, the hosting stack is architected so that building and operating the server directly strengthens `qualia-core-db` and `qualia-client-core`. All engine, logic, storage, and networking primitives reside in the QualiaDB crates, while the administrative UI and client surfaces reside in Poet and `webizen-browser`.

2.1 Dual Deployment Modes: Standalone VPS & Local Desktop Extension

- **Mode A — Standalone Sovereign VPS Host (`qualia-cli host init --profile=vps`):** Installs directly on any modern Linux VPS (Debian, Ubuntu, Alpine, or NixOS on x86_64 or ARM64; 2 vCPU / 4 GB RAM baseline, scaling to dedicated NVMe/ZNS storage and optional Vulkan/WebGPU compute adapters per Optimizing WGPU for AI and QualiaDB). Runs the native daemon (`services/daemon.rs` on `:4242` and HTTPS/QUIC `:443`), the authoritative `net/dns/` server on `:53`/DoH/DoQ, the `qualia-solid-bridge` LDP pod server, and the cold-path ontology ingestion worker.
- **Mode B — Online Extension Relay for Local `webizen-desktop` / `qualiadb`:** Pairs cryptographically with a personal or family `webizen-desktop` instance (running locally on Tauri `:8080` per Webizen Desktop Application Requirements) via `crates/qualia-peer` (`PeerHost::pair`, `exchange_protected`) using post-quantum `ML-KEM-768` handshakes and `Ed25519` signatures over `SocialWebNet` (userspace WireGuard) or `CSCP` WSS/HTTP2 capsules.
- **Strict "Church & State" Vault Separation:** Following the architectural principle established in GuideForDevelopers, every deployment enforces a cryptographic boundary between the **"Church" Database (`TheWebizenAgent` / Sanctuary Keep)**—which holds deeply personal biographical, health (`wellfare-core`), and unshared consciousness records strictly on local hardware to protect *Freedom of Thought*—and the **"State" Database (`PermissiveCommonsTech`)** hosted on the VPS, which holds shared social records, public websites, and `ValuesCredentials`-governed ontologies. If the VPS stores off-site disaster-recovery backups of the local vault, they are transmitted solely as client-encrypted `.q42j` journal segments recoverable only via the local CLI utility with M-of-N Shamir secret shares.

2.2 How Building `webizen-host` Directly Improves the QualiaDB Stack

Auditing the QualiaDB Capability Reference (0.0.40) reveals several complete or partial engine modules that currently lack a live production caller or public network qualification. Implementing `webizen-host` closes six concrete repository gaps as a direct consequence of the hosting work:

1. **Qualifying `CSCP-08` & `CSCP-09` on the Public Internet:** As documented on webizen.au/qdnf.html, the `CSCP-08` (public Internet WSS/MASQUE bound-UDP relay) and `CSCP-09` two-host evidence flags remain `false` solely because GitHub Pages is static and a live second host URL controlled by the operator is required. Deploying `webizen-host` provides the live public endpoint to complete `CSCP-08/09` qualification.
2. **Completing `FLAG_PAYLOAD_CATALOGUE` (`0x0080`, ADR 0014) in `q42_volume.rs`:** Upgrades the `.q42` volume payload catalogue from stub status to full read/write parity with `bundle/cbor_ld_manifest.rs` (`PayloadCatalogueEntry`).
3. **Wiring `owl/materialize.rs` & `owl/shacl_convert.rs` into Production Ingest:** Connects the complete but currently *Library-only* OWL 2 RL forward-chaining materializer and OWL-to-SHACL converter into the VPS ontology build pipeline.
4. **Replacing `mock_network_fetch` in `net/fetch_10d.rs`:** Replaces the mock transport wrapper with real content-addressed range fetching over QPR and HTTP/2 while preserving the existing consent and CRC-32C verification

gates.

5. **Hardening Multi-Tenant Key Derivation (`identity/agency.rs`):** Upgrades `derive_lane_key` from single-pass SHA-256 to memory-hard Argon2id + HKDF using `specialized_libs/cryptographic_library`.
6. **Binding Persistent Commons Pools to Asset Manifests:** Connects the zero-heap arithmetic in `modalities/value_flow.rs` to durable `.q42j` transaction journals on the VPS so `is_commons_discharged()` automatically triggers license transitions in production.

3. Large Ontology Ingestion, Processing & Segmented Hosting (`>100 MB WebCivics/ns`)

The immediate catalyst for this specification is hosting and processing https://github.com/WebCivics/ns (which exceeds 100 MB and cannot be imported into standard browser or cloud tools) alongside the 102 UN/OHCHR/ICRC human-rights instruments in `core-ontologies/`, Australian Bureau of Statistics (ABS) datasets, medical/chemical ontologies (SNOMED, ChEBI, QUDT, DICOM/RadLex), and Ecosphere Assets species corpora. When ontologies are hosted as raw monolithic RDF/XML or Turtle files, clients must download tens or hundreds of megabytes just to resolve a single class or rule, making mobile execution impossible and driving up server egress costs.

3.1 Streaming Cold-Path Ingestion & Compilation Pipeline

On `webizen-host`, ontology compilation runs as a supervised background service governed by QualiaDB's Tier 2 `ColdBounded` allocation contract (`AGENTS.md` §0-A, defaulting to a strict `256 MiB` arena budget that fails closed on exhaustion rather than crashing the host OS):

- **Step 1 — Repository Sync & Provenance Hashing (`q42/source_catalogue/` & `q42/asset_import/`):** The host daemon clones or pulls `https://github.com/WebCivics/ns` (and configured upstream Git/HTTP mirrors) into a content-addressed staging directory, recording git commit tree hashes, BLAKE3 file digests, and upstream license obligations in `q42/asset_envelope/` (where unknown licenses fail closed and derived volumes inherit the union of upstream obligations).
- **Step 2 — Streaming Parse & `Q42LEX v4` Namespaced Lexicon Interning (`query/ingest*.rs`, `q42_lex_ns.rs`):** Rather than loading a 150 MB DOM tree into RAM, the streaming parser ingests RDF/XML, Turtle, N-Triples, JSON-LD, YAML-LD, and Notation3 (`=>` strict, `~>` defeasible, `^>` defeater, `-o` linear rules) in bounded byte chunks. It enforces `IngestMode::Complete` (never `StripLiterals`) and writes terms into the **v4 namespaced paged lexicon (`Q42LEX v4`)**, which deduplicates URI prefixes across the entire `WebCivics/ns` corpus while computing 64-bit FNV-1a `q_hash()` identifiers for every IRI.
- **Step 3 — SHACL-First Non-Thing Uplift & Bytecode Compilation (`owl/shacl_convert.rs`, `governance/webizen_bytecode.rs`):** Enforces the normative rule in `docs/manuals/standards/shacl-first-vs-owl-ok-class-list.md`: *Natural persons, kinship, human relations, and living ecological entities are rdfs:Class + SHACL/ShEx and must never subclass `owl:Thing`.* Legacy OWL definitions inside imported repositories are automatically uplifted into W3C SHACL shape packs (`shapes/*.shacl.ttl`) and compiled into WAM-style `SlgOpcode` bytecode (deontic `0x10–0x12`, epistemic `0x20–0x22`, paraconsistent `0x30–0x32`, LTL `0x40–0x44`).

Compiled Artifact Format	Internal Binary Layout & Alignment	How It Transforms >100 MB `WebCivics/ns`	Mobile, Browser WASM & Peer Access Method
`.q42` v4 Semantic Volume (`q42/q42_volume.rs`)	64-B header + 40,960-B LZ4-compressed `QualiaSuperBlocks` (853 × 48-byte `NQuins`) + `BIDX` block index + postings index + `Q42LEX v4` paged lexicon.	Compresses verbose RDF/XML & Turtle by 6–12× into fixed 48-byte `NQuin` quads sorted by subject/predicate/context for zero-copy mmap and GPU BDA traversal.	Clients fetch the 64-B header + `BIDX` index and request **only single 40 KiB SuperBlocks** via `range_volume.rs` (HTTP `206` or `QPR` slice).

Compiled Artifact Format	Internal Binary Layout & Alignment	How It Transforms >100 MB `WebCivics/ns`	Mobile, Browser WASM & Peer Access Method
`QBDL .hmc` Hypermedia Bundle (`bundle/format.rs`)	Magic `QBDL` v1, 64-B header, 64-B page-aligned intact entries, trailing CBOR index footer (`offset`, `length`, `sha256`), whole-file CRC-32C + `cbor_ld_manifest.rs`.	Packages domain-sharded `.q42` volumes, signed `law_packages.rs`, `.10d` manifolds, and multi-language `PayloadCatalogueEntry` attachments into one verifiable archive.	`entry.offset + seg.offset` enables zero-copy range extraction of interior sub-ontologies without downloading the whole bundle.
`.vibe` / `VBC1` Vocab Chunks (`vibe/src/vocab/`)	Tree-shaken `.n3` vocabulary chunk tables + golden CBOR-AST hashes and `VBC1` bytecode chunks with `u64` execution budgets.	Extracts high-frequency IRIs (e.g., human-rights predicates, SNOMED, QUDT, WebCivics terms) into compact lookup tables with zero string-table bloat.	Directly imported into VibeScript 0.1 and the 640 KiB `Ontology MCP` WASM kernel.

3.2 Content-Negotiated Virtual Namespace Server (Upgrading `ns.webcivics.net`)

Once `WebCivics/ns` and `core-ontologies/` are compiled into memory-mapped `.q42` v4 volumes on the VPS, `webizen-host` exposes a zero-allocation **Virtual Namespace Handler** that serves any URI under `https://ns.webcivics.net/*` or hosted custom domains via HTTP content negotiation:

- **Qualia / Webizen Native Clients (`Accept: application/q42` or `application/cbor-ld`):** Receives the exact 48-byte `NQuin` slice, `Q42LEX v4` token page, and Ed25519-signed provenance receipt directly from the memory-mapped `QualiaSuperBlock` in sub-millisecond latency.
- **Legacy Semantic Web & W3C Clients (`Accept: text/turtle`, `application/n-triples`, `application/ld+json`, `application/rdf+xml`):** Dynamically reconstructs standard RDF serializations on the fly from the `.q42` volume using `query/resolver.rs::format_ntriples_to` and the paged lexicon—eliminating the need to store duplicate text files on disk.
- **Human Browser Visitors (`Accept: text/html`):** Serves a lightweight, interactive HTML + webizen-lite-wasm (640 KiB) ontology explorer that renders the term's definition, SHACL shape constraints, deontic modality rules, and interactive graph neighborhood entirely in the visitor's browser.

4. Ownership, Identity & Multi-Person Stewardship Architecture

The user's specification requires that a server *"should either be owned by a group (many natural persons) or a natural person, it may act to extend local services to an online location to support personal info, or be used to support community / group resources, etc."* In QualiaDB, this requirement is enforced at the cryptographic and logical root of the operating system.

4.1 The Foundational Invariant: Who-Plane (`NaturalAgent`) ≠ Handle-Plane (DNS / ABN)

Conventional hosting panels (cPanel, Cloudflare, AWS) conflate identity with a DNS domain name or a corporate billing account. As established on webizen.au/qdnf.html, `webizen-host` enforces a strict separation between the **Who-Plane (Identifier Fabric)** and the **Handle-Plane**:

- **Natural Persons on the Who-Plane:** Every human owner, family member, or community co-steward is represented as a `NaturalAgent` authenticated via self-certifying `did:qi` (`crates/qualia-core-db/src/did_qi/` signed `QCDE-1` lifecycle vectors) and `WebizenID` keys stored in their hardware/local `key_vault.rs`.
- **Domains & Registries on the Handle-Plane:** Domain names (`ns.webcivics.net`, `civics.au`), IPv4/IPv6 endpoints, expiring `DNI` locators, and state-issued registry strings (ABNs, company numbers) exist solely as attachable/detachable handles.
- **Collapse-Detector Enforcement (`Handle revoke ≠ who-erase`):** If an ICANN domain name expires or is seized by a registrar, the host's `Collapse Detector` transitions the domain locator to `held/closed` while keeping all `NaturalAgent`

identities, cooperative memberships, and `.q42` / Sanctuary Keep volumes 100% intact and reachable over native QDNF (`QResolve`) or any newly attached domain handle.

Server Ownership Archetype	Cryptographic & Jural Control Model	Primary Hosting & Extension Use Cases	Implementing QualiaDB Crates & Modules
A. Single Natural Person (Individual Sovereign Host)	1-of-1 root `NaturalAgent` (`did:qi`) authority + delegated device sub-keys (desktop, laptop, phone companion PWA) via `ZCAP-LD` (`modalities/delegation.rs`).	Online extension for local `webizen-desktop`; personal domain & sovereign mail; encrypted off-site `.q42j` vault backup; personal apps/blogs.	`identity/webizen_did.rs`, `did_qi/`, `governance/consent_contract.rs`
B. Family / Household Collective (Shared Domain + Guardianship)	1–2+ adult `NaturalAgent` stewards + dependent profiles governed by `GuardianshipRelations` / `GuardianshipSemantics` and capacity rules (`modalities/capacity.rs`).	Shared family domain (`family.tld`) with isolated per-member Sanctuary Keep lanes (Argon2id `derive_lane_key`), child-safe sandboxes, and elder/dependent care records.	`wellfare-core/src/guardianship.rs`, `modalities/capacity.rs`, `identity/agency.rs`
C. Group / Community / Cooperative (Many Natural Persons)	**M-of-N Steward Quorum** over a 32-slot `SuspendedTransactionQueue` (`foundation/crdt.rs`) + BFT consensus (`f=⌊(n-1)/3⌋`) and Hohfeldian jural checks (`modalities/jural.rs`).	Community & civic hubs (ns.webcivics.net, web.civics.au), humanitarian foundations, scientific ontology commons, and cooperative ISP/CDN nodes.	`qualia-cooperative-core`, `domains/geospatial/stewardship.rs`, `governance/coordination.rs` (`0x70–0x7F`)

4.2 Cooperative & Multi-Person Governance Mechanics (`qualia-cooperative-core`)

When a server is owned by a group of natural persons (such as a WebCivics faculty, an Indigenous knowledge stewardship circle enforcing Traditional Knowledge / Biocultural labels via `governance/tk_bc_labels.rs`, or a neighborhood micro-grid cooperative), administrative actions are governed by `crates/qualia-cooperative-core` and the Sentinel VM's multi-agent coordination opcodes (`0x70 OP_AUTHORIZATION_GRANT`, `0x71 OP_RESOURCE_DECLARATION`, `0x72 OP_PERFORMANCE_RATING` in `governance/coordination.rs`):

- **Quorum-Gated Server Mutations:** Privileged operations—such as modifying domain DNS zones, admitting a new canonical ontology volume, changing monthly egress cost caps, or disbursing accumulated micropayment pools—cannot be executed by a single compromised admin key. Instead, the request is written as a pending `NQuin` transaction into the 32-slot `SuspendedTransactionQueue` (`foundation/crdt.rs`) and executes atomically only after `M-of-N` designated stewards sign the transaction with an explicit-denial guard (`domains/geospatial/stewardship.rs`).
- **Paraconsistent Conflict Retention (`governance/illocution.rs`):** If co-stewards submit conflicting policy rules (governed by `core-ontologies/modal-junctures.n3`), the engine uses paraconsistent logic (`0x30–0x32`) and Hohfeldian correlative-gap detection (`modalities/jural.rs`) to retain both claims and flag `RequiresHumanReview` rather than silently overwriting a steward's dissent.

5. cPanel-Replacement Hosting Stack & Three-Tier Sandboxing

To replace cPanel, Plesk, and Docker-heavy control panels completely, `webizen-host` provides a unified hosting environment where every website, domain, ontology service, or community application runs inside a deterministic **Sandbox Cell (`net/peer/cells/` + `governance/cell_graph.rs`)** managed through a human-centric administrative interface.

5.1 Three-Tier Sandbox Architecture for Hosted Services & Websites

Rather than relying on heavyweight full-OS virtual machines or insecure shared scripting runtimes, `webizen-host` isolates workloads across three complementary sandbox layers:

1. **Sandbox Tier 1 — VibeScript (`VBC1`) & 42 MiB `SlgArena` Cellular Sandbox (Native Semantic Apps):**
   - Replaces server-side Node.js/V8 and PHP with VibeScript 0.1 (`crates/vibe/`)—a typed, capability-bounded, non-JIT language compiled to `VBC1` bytecode.
   - **Hard Cycle & Memory Bounds:** Every function in a `VBC1` chunk carries an explicit `u64` execution budget (`vibe/src/budget.rs`) and is constrained by the effect lattice (`Pure < Hot < Cold < Async < External` in `vibe/src/effects.rs`). Hot-path graph queries and N3/SHACL rules execute inside an isolated **42 MiB `SlgArena` Sentinel cell** (`governance/cell_graph.rs`) linked by `PermissiveRoutingLane` channels, guaranteeing zero heap allocation (`HotZeroHeap`) and preventing runaway scripts from affecting sibling sites.
   - **Fail-Closed Capability Leases:** A sandboxed script cannot touch the filesystem, network, or ~1,158 host catalog operations (`vibe/src/catalog/ids.rs`) unless the sandbox manifest explicitly grants a `using {Family};` lease verified at runtime by `deontic_interrupt.rs`.
2. **Sandbox Tier 2 — WebAssembly & QApp Isolate Sandbox (Portable Client/Server Apps):**
   - Executes compiled Rust/WASM application modules (conforming to the source-independent `SDK-01..10` contract in `qualia-app-contract`), `vibe-wasm` packages, and the 640 KiB `webizen-lite-wasm` Ontology MCP endpoint.
   - Each QApp (`q42/app_manifest/`, `q42/app_registry/`) mounts only its own scoped virtual volume prefix (`wasm_storage.rs`) and communicates with the host strictly through typed command/event buffers.
3. **Sandbox Tier 3 — W3C Solid LDP Pod Bridge & Linux/eBPF Container Sandbox (Web 2.0 & Interop Sites):**
   - **Native W3C Solid Pod Hosting (`crates/qualia-solid-bridge`):** Incorporates the MIT-licensed Solid/LDP baseline deployed at web.civics.au, providing per-user and per-group Solid Pods with a strict Turtle↔`NQuin` translation boundary so users can run standard Solid apps or migrate their data out of QualiaDB at any time.
   - **Linux Namespaces + eBPF Socket Perimeter:** Static HTML5/CSS sites, reverse-proxied legacy apps, and media processors run inside unprivileged Linux user, mount, PID, and network namespaces guarded by `qualia-core-db`'s native **eBPF filter/firewall (`net/`)**, enforcing per-sandbox bandwidth ceilings, CPU cgroup quotas, and read-only root mounts.

5.2 cPanel-Parity Feature Matrix & Human-Centric Admin UX

Addressing the usability requirements in UI / UX / Human Centric Systems Considerations, the server control panel is designed for **human beings first**—organized around Webizen's six life domains (Memory · Relations · Care · Practice · World · Instruments) inside the Poet / Dioxus workbench (with first-class WASM support on desktop and mobile):

cPanel / WHM Feature	Webizen-Host Equivalent Subsystem	Human-Centric UX & QualiaDB Engineering Implementation
Domain, Subdomain & DNS Zone Editor	**World & Realm Domain Manager**	Visual onboarding wizard to attach ICANN domains/subdomains, auto-configure NS/glue records, bind `did:qi` handles, and toggle 1-click decentralized CDN seeding.
SSL/TLS Manager & AutoSSL	**WebizenID-TLS & ACME Authority**	Automates Let's Encrypt TLS certs for standard browsers while issuing **WebizenID-TLS** client certificates (with SAN bound to the user's Webizen RDF/DID endpoint) and `Groth16 over BLS12-381` (`crypto/`) zero-knowledge standing proofs.
Email Accounts, Forwarders & Webmail	**Relations Sovereign Mail & Relationship Aliases**	Full custom-domain email (`@yourdomain.tld`) backed by `qualia-client-core/src/mail` and `did:qi` mailbox binding. Implements **relationship-scoped email delegation** (delegating a distinct address per contact/relationship, whitelisted via N3 ontology rules to eliminate spam and correlation).

cPanel / WHM Feature	Webizen-Host Equivalent Subsystem	Human-Centric UX & QualiaDB Engineering Implementation
Databases (MySQL / phpMyAdmin)	**QualiaDB Volume, Ontology & SPARQL/N3 Studio**	Manage `.q42` v4 volumes, `.q42j` journals, and 1-click Git ontology imports (e.g., `WebCivics/ns`). Includes visual SPARQL-Star, N3 rule authoring, and SHACL validation panels.
File Manager, FTP & Media	**Hypermedia Bundle (`.hmc` / `.10d`) & Solid Explorer**	Drag-and-drop management of `QBDL` `.hmc` bundles, `.10d` 3D/10D spatial assets, and Solid Pod folders with automatic EXIF/WAV provenance sidecars.
Metrics, Logs & Billing Controls	**Hard Cost-Cap & Commons Value-Flow HUD**	Real-time telemetry of bandwidth, energy (Joules), and compute per sandbox with a physical-style slider to lock in a **hard monthly dollar/egress ceiling** alongside live CDN micropayment earnings.

6. QualiaDB-Accelerated DNS, Naming & QDNF Routing

A core requirement of the prompt is leveraging `qualiadb` capabilities to improve DNS performance and related network resolution. Inside `crates/qualia-core-db/src/net/dns/` (wire codec, resolver, cache ring, quin records, ontology, SDN, lossless archive) and `net/qdnf/`, QualiaDB replaces string-heavy legacy DNS parsing with fixed-width 48-byte `NQuin` lookup rings and cryptographic routing.

6.1 How QualiaDB Accelerates DNS Resolution & Resilience

- **O(1) Compile-Time & Ingest-Time `q_hash()` Resolution over 48-Byte `NQuin` Rings:**
  - In legacy DNS servers (BIND, Unbound, CoreDNS), every incoming packet triggers variable-length label parsing, case-folding, and pointer-chasing across heap-allocated radix trees.
  - In `webizen-host`, authoritative zones and cached recursive records are stored as **48-byte `NQuins`** in a pre-allocated, cache-line-aligned `HotZeroHeap` ring buffer (`net/dns/`). The domain label + RRType is hashed via 64-bit FNV-1a (`q_hash()`) into the `subject` slot; routing policy and TTL epoch pack into `predicate/metadata`; and IPv4/IPv6 addresses, `DANE` TLSA hashes, or `did:q42` lexicon pointers pack directly into the 64-bit `object` slot.
  - Combined with optional Linux **AF_XDP / eBPF** kernel-bypass packet processing (`net/`), standard `A`, `AAAA`, `HTTPS`, and `TXT` queries resolve in **O(1) constant time with zero heap allocations**, sustaining millions of queries per second on a modest VPS while resisting DNS amplification floods.
- **In-Flight Deontic & SHACL Policy DNS Firewall (`QPolicy`):**
  - Because DNS records are native `NQuins`, the resolver can evaluate N3/deontic policy rules in the 42 MiB `SlgArena` and check `net/qdnf/policy_labels/` (Information Flow Control lattice) before returning an answer.
  - This enables **Consent-Gated Split-Horizon DNS** (returning private device/vault IPv6/ULA records only when the requester presents a valid `DNI` cookie or DoH/DoQ `WebizenID-TLS` session, and returning `NXDOMAIN` to public scrapers) as well as automated blocking of known malicious or jurisdictionally non-compliant telemetry endpoints.
- **Bitemporal Lossless DNS Archive (`net/dns/` lossless archive):**
  - Every DNS zone change and DNSSEC/DANE key rollover is recorded in an append-only `.q42j` journal using bitemporal predicates (`lifecycle:validFrom`, `lifecycle:transactionFrom`), creating a cryptographic evidentiary chain of domain stewardship.

6.2 Dual-Stack Integration with the Qualia Decentralized Network Fabric (QDNF)

Every domain hosted on `webizen-host` bridges the Legacy Internet Gateway with the six-layer QDNF Protocol Honeycomb:

- **`QLink` & `DNI` (Cryptographic Adjacency & Expiring Locators):** Replaces spoofable ARP/BGP assumptions inside the Webizen mesh with proven neighbour adjacency and expiring self-certifying topological locators (`DNI`).

- **`QRoute` & Cryptokey IPv6 (`governance/web_civics.rs`):** Implements constraint-first routing (`validate → index → plan` in `net/qdnf/route/`) paired with deterministic **Ed25519 → ULA IPv6 derivation**, where an agent's cryptographic key mathematically derives its overlay IPv6 address across `SocialWebNet` (WireGuard) and `CSCP` transition bearers.
- **`QResolve` (`net/qdnf/resolve/qsr`):** Resolves `did:qi`, `did:q42`, and content-addressed `.q42`/`.hmc` digests directly across peer nodes so that even if ICANN DNS is unreachable or censored, Webizen clients resolve hosted domains and ontologies seamlessly.

7. User-Powered Decentralized CDN & Hard Cost Caps (Solving `ns.webcivics.net`)

To permanently solve the hosting cost crisis experienced on ns.webcivics.net—where heavy traffic from two major US hubs risked uncapped AWS CloudFront egress bills—`webizen-host` transforms every hosted domain into a self-capping origin backed by a **User-Powered Decentralized CDN (`Q-CDN`)**.

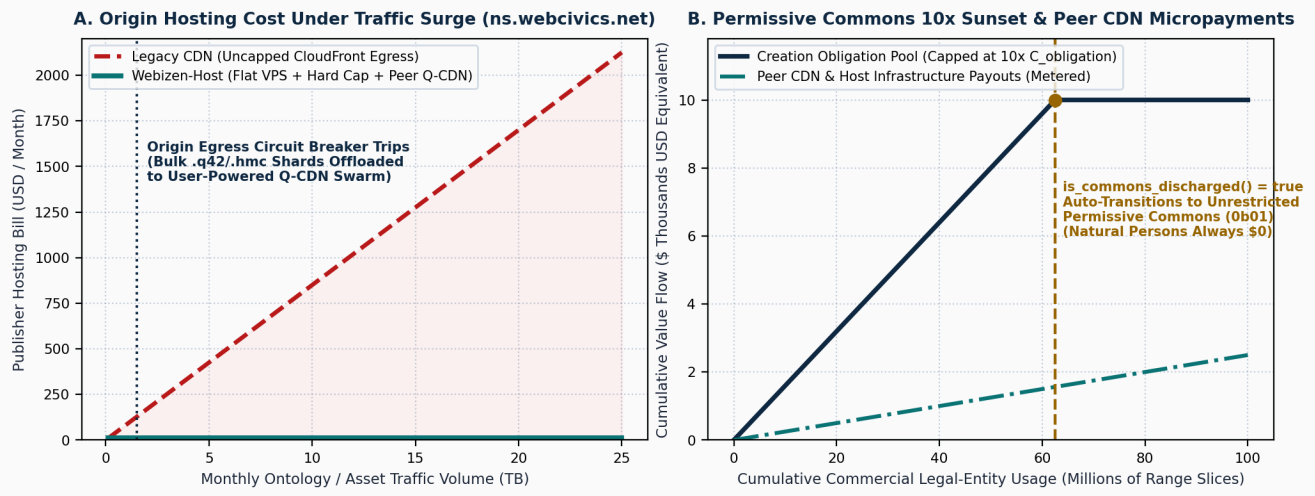


Figure 2: *Left (Panel A):* Comparison of monthly publisher hosting cost exposure under a traffic surge (0.1 to 25 TB/month), showing how `webizen-host`'s hard egress circuit breaker caps origin cost at the flat VPS rate while offloading bulk shards to the user-powered Q-CDN swarm. *Right (Panel B):* The two-loop Permissive Commons economic curve (`modalities/value_flow.rs`), illustrating commercial toll accumulation up to the `10×` Obligation Cost sunset alongside continuous metered peer CDN payouts.

7.1 Architecture of the User-Powered Decentralized CDN (`Q-CDN`)

Because QualiaDB compiles large ontologies and media into **content-addressed, chunk-verifiable containers**—specifically 40,960-byte LZ4 `QualiaSuperBlocks` in `.q42` v4 and 64-byte page-aligned entries with SHA-256 footer digests in `QBDL .hmc` bundles—any peer can serve cached slices without the requester having to trust the peer:

1. **Origin Serves Lightweight Signed Root Index (<10 KiB):** When a client requests a resource from `ns.webcivics.net` or any hosted domain, the origin VPS serves the compact, Ed25519-signed **Master Index** (`cbor_ld_manifest.rs` / `BIDX` header), which lists the SHA-256 / BLAKE3 digest, byte offset, and micropayment weight of every 40 KiB SuperBlock or `.hmc` segment.
2. **Three-Layer Peer Swarm Distribution (`QPR` + WebTorrent `:4242` + Browser OPFS):**
   - **Layer A — Cooperative VPS Mirrors (`net/peer/replication/`):** Other `webizen-host` VPS instances participating in the Webizen Alliance replicate hot `.q42` and `.hmc` shards using Merkle custody proofs, tombstones, and resumable transfers, advertised via DNS `HTTPS`/`SVCB` and HTTP `Alt-Svc` headers.
   - **Layer B — Desktop User Nodes (`webizen-desktop` via `qualia-peer`):** Every user running `webizen-desktop` caches the `.q42` ontology SuperBlocks and `.hmc` assets they use locally and seeds verified slices to peers over `QSession/QSync` and the built-in daemon seeder (`services/webtorrent_*.rs` on port `:4242`).
   - **Layer C — Browser Visitors (OPFS Cache + WebRTC/WebTorrent):** Browser visitors using the WASM portal cache fetched 40 KiB SuperBlocks in their Origin Private File System (OPFS, `wasm_storage.rs`) and exchange verified chunks with fellow browser peers via WebRTC DataChannels.
3. **Hard Origin Egress Circuit Breaker (`net/qdnf/economics/meters.rs`):**

- The server owner sets a strict byte and dollar ceiling in `meters.rs` (e.g., *"Max 1.5 TB/month direct origin egress; $0.00 permitted overage"*) enforced by the Sentinel VM's anti-usury circuit breaker opcodes (`governance/coordination.rs`).
- When traffic from US data hubs or scrapers pushes origin egress toward the threshold, the circuit breaker trips automatically: the VPS continues serving the tiny signed Root Index and verified Humanitarian ICT sessions directly, while redirecting all bulk `.q42`/`.hmc` range fetches (via `HTTP 307`, `Link: rel="duplicate"`, and `QResolve` swarm pointers) to the user-powered `Q-CDN` swarm or commercial toll-gated mirrors.

8. Micropayments, Permissive Commons Economics & Blockchain / DLT Plan

To fund the ongoing curation of ns.webcivics.net and reward users who contribute bandwidth and storage to the decentralized CDN, `webizen-host` wires the already-implemented **Permissive Commons economic engine (`crates/qualia-core-db/src/modalities/value_flow.rs`)** and **QDNF metering substrate (`net/qdnf/economics/meters.rs`)** to real-time micropayment and DLT settlement rails.

8.1 Two-Loop Permissive Commons Economic Model (`value_flow.rs`)

- **Loop 1 — Capital Creation Recovery (`Obligation Cost` & `10×` Commons Sunset):**
  - Processing and curating large ontologies (`WebCivics/ns`) or Ecosphere Assets incurs verifiable engineering and compute costs recorded in `cbor_ld_manifest.rs` as the baseline **Obligation Cost ($C_{obligation}$)**, with a capped recovery ceiling of up to **10× $C_{obligation}$** computed via `commons_cost(production_cost, roi_cap_percent, max_roi_percent)` and guarded by `check_usury()`.
  - **Humanitarian ICT & Natural Person Exemption:** Requests carrying a verified `NaturalAgent` or `humanitarian-ict.n3` credential receive a `0%` royalty multiplier (`royalty()` = 0), guaranteeing free access for citizens, researchers, and human-rights defenders.
  - **Commercial Legal-Entity Toll Gate (HTTP `402 Payment Required` / `QPolicy`):** Corporate scrapers, commercial AI training hubs, and incorporated legal entities must attach an ILP-STREAM or eCash micropayment proof per range request. Payments accumulate via `pool_after()` and `accumulate_meter_to_pool()`.
  - **Automatic Commons Sunset:** The instant cumulative commercial tolls reach the capped obligation cost, `is_commons_discharged(pool, cost)` evaluates to `true` and invokes `CommonsObligation::discharge_to_commons()`, permanently transitioning the asset to `PermissiveRoutingLane::EnforcePermissiveCommons` (`0b01`) for all users worldwide—eradicating perpetual rent extraction ("Ending Digital Slavery").
- **Loop 2 — Ongoing Infrastructure Micropayments for Peer CDN Seeders & Hosts:**
  - Independent of creation recovery, serving bandwidth and running remote SPARQL/inference jobs consumes live resources measured by `net/qdnf/economics/meters.rs` (`MeterKind::{Bytes, EnergyJoules, Time, Compute}`).
  - `value_flow.rs::carriage_fee_from_meters()` calculates deterministic micro-tariffs per verified 40 KiB SuperBlock or `.hmc` range slice delivered, routing the payout directly to the **user nodes (`webizen-desktop` seeders)** and **independent VPS hosts** that served the bytes.

8.2 Blockchain & DLT Architecture: Forking vs. Native Hybrid Evaluation

The prompt asks how to support a blockchain solution—*"whether it be by forking an existing solution or otherwise."* A source audit of `qualia-core-db` and `qualia-client-core` shows four powerful ledger and payment components already implemented in-tree: (1) `services/ilp_dispatcher.rs` (real **ILP-STREAM-over-SPSP** micropayment dispatcher writing every payment as a signed N-Quad receipt into `.q42`, with USDC/XRPL fallbacks); (2) `qualia-client-core`'s **BIP32/secp256k1 HD wallet** with native **eCash (XEC) integration via `chronik.rs`**; (3) `did_qi/` **UTXO anchoring**; and (4) `modalities/consensus.rs` (**BFT consensus `f=⌊(n-1)/3⌋`**) + `foundation/crdt.rs`. Rather than choosing between a slow greenfield blockchain or an unintegrated external chain, we specify a **Three-Tier Sovereign DLT Architecture**:

DLT / Ledger Architecture Tier	Technical Mechanism & Codebase Integration	Strengths & Trade-Offs	Deployment Phase in `webizen-host`
Tier 1: eCash (XEC / `chronik.rs`) UTXO Base + ILP-STREAM Off-Chain Micropayments *(No-Fork Immediate Production Rail)*	Uses the in-tree `chronik.rs` indexer + BIP32 wallet for sub-cent (~$0.0001), <2s Avalanche-finalized UTXO settlements and `did:qi` anchoring, while `services/ilp_dispatcher.rs` streams high-frequency per-chunk CDN micropayments and logs signed N-Quad receipts into `.q42j`.	**Strengths:** 100% implemented in-tree today; near-zero energy/fee footprint; immediate revenue for `ns.webcivics.net`. **Trade-Off:** Base UTXO asset is external XEC/stablecoin rather than a native protocol coin.	**Phase 1 (Immediate Launch):** Turns on commercial toll collection and peer CDN payouts on Day 1 with zero consensus maintenance.
Tier 2: Native QDNF DAG-CRDT + BFT Committee Clearinghouse *(Blockless Local/Realm Ledger)*	Uses append-only `.q42j` hash-chained N-Quad journals, `modalities/reconciliation.rs` (`compare_views`), and realm **BFT committees (`modalities/consensus.rs`, `f=⌊(n-1)/3⌋`)** across cooperative VPS hosts to clear bilateral CDN bandwidth credits and track `commons_cost` pools without mining blocks.	**Strengths:** Native 48-byte `NQuin` ABI; zero gas fees for intra-realm credit netting; works offline/partitioned. **Trade-Off:** Needs periodic global checkpoint anchoring across untrusted realms.	**Phase 1.5 (Realm Accounting):** Runs inside every `webizen-host` cluster to meter bytes/Joules and batch net payouts onto Tier 1 or Tier 3.
Tier 3: Forked eCash/Avalanche-UTXO or Rust BFT Sovereign Chain ("QualiaChain") *(Sovereign Fork Option)*	Forks the open-source **eCash (ABC/Avalanche-UTXO + Chronik Rust indexer)** or a lightweight **Rust BFT engine** and embeds **`qualia-core-db`'s 42 MiB `SlgArena` VM (`governance/webizen.rs`)** directly into block/tx validation—so L1 consensus natively enforces `value_flow.rs` (`check_usury`, `10×` commons sunset, and `NaturalAgent` fee exemptions).	**Strengths:** Every `webizen-host` VPS acts as a sovereign validator; protocol economics and human-rights deontic rules are enforced at L1 consensus; preserves existing `chronik.rs` & `did:qi` UTXO code! **Trade-Off:** Requires 15–50+ independent VPS hosts for decentralized validator security.	**Phase 2 (Webizen Alliance Mainnet):** Launched by forking the Chronik/Avalanche-UTXO stack once the initial cohort of community VPS hosts is online.

Why Forking the eCash / Chronik + Avalanche-UTXO Stack is the Optimal Fork Path: Because `qualia-client-core` already implements BIP32/secp256k1 P2PKH and the `chronik.rs` eCash indexer, and `did_qi/` already implements UTXO anchoring, forking an eCash/Chronik-compatible UTXO node (augmented with deterministic Avalanche pre-consensus and an embedded Rust `SlgArena` validation hook for `CommonsObligation` Covenants) requires zero rewrites of the client wallet or identity layer while giving the Webizen ecosystem its own energy-efficient, non-usurious sovereign ledger.

9. Phased Implementation Plan & Work Packages (WP-01 to WP-08)

To deliver a working, installable VPS platform that immediately solves the WebCivics/ns (>100 MB) and ns.webcivics.net hosting challenges while building out the full cPanel-replacement stack, execution is organized into eight sequential Work Packages mapped directly to the `qualia-core-db`, `qualia-client-core`, `vibe`, and `poet` crates. In accordance with `AGENTS.md` §0-A/B and the source-independent SDK release gate (`GA-17` / `SDK-01..10` in the QualiaDB Capability Reference), each work package closes only when its complete end-to-end chain—*producer → serialized container (`.q42` / `QBDL .hmc` / `VBC1`) → bounded admission → runtime caller → verified acceptance receipt*—is demonstrated on both a live Linux VPS target and a paired `webizen-desktop` client.

Work Package	Target Crates & Modules	Engineering Deliverables & Integration Tasks	Acceptance & Exit Evidence
WP-01 (P0) Large Ontology Streaming Compiler	`qualia-core-db/src/q42/`, `q42_lex_ns.rs`, `qualia-cli/src/ingest/`	1. Configure `q42/asset_import/` streaming worker to pull & chunk `https://github.com/WebCivics/ns` (>100 MB). 2. Enforce `IngestMode::Complete` + `Q42LEX v4` paged lexicon + `owl/shacl_convert.rs` (non-Thing uplift). 3. Complete `FLAG_PAYLOAD_CATALOGUE` (`0x0080`) writer in `q42_volume.rs` and emit `QBDL .hmc` bundles.	Full `WebCivics/ns` repo compiles within a `256 MiB` cold memory budget with zero OOMs; `verify-graph --encoder=import` passes 100%.
WP-02 (P0) Range-Served Virtual Namespace & Cost Cap	`services/daemon.rs`, `q42/volume/range_volume.rs`, `net/qdnf/economics/meters.rs`	1. Replace static file serving on `ns.webcivics.net` with zero-alloc HTTP/2 & HTTP/3 content-negotiated handler (`.q42` slice, Turtle, JSON-LD, HTML+WASM). 2. Wire `meters.rs` hard monthly/daily egress byte & dollar circuit breaker.	`ns.webcivics.net` serves 40 KiB `QualiaSuperBlock` range queries in <2 ms; simulated US-hub traffic surge trips the hard cap cleanly with $0 overage.
WP-03 (P0/P1) QualiaDB `NQuin` DNS & QDNF Relay	`net/dns/`, `net/qdnf/resolve/`, `crates/qualia-peer`	1. Expose `net/dns/` 48-byte `NQuin` zone ring on UDP/TCP `:53`, DoH, DoT, and DoQ with automatic DNSSEC/DANE & `_did` records. 2. Use the live VPS hostname to execute two-host Internet trials for **CSCP-08** (MASQUE bound-UDP) and **CSCP-09**.	O(1) `q_hash()` DNS benchmark verified; `CSCP-08` and `CSCP-09` public Internet evidence flags on webizen.au/qdnf.html flip to verified.
WP-04 (P1) Three-Tier Multi-Tenant Sandbox	`crates/vibe/`, `governance/cell_graph.rs`, `qualia-solid-bridge`	1. Build `HostSandboxManager` isolating tenants into 42 MiB `SlgArena` cells + VibeScript `VBC1` `u64` cycle budgets. 2. Wire WASM QApp isolates, `qualia-solid-bridge` (web.civics.au pods), and Linux cgroup/namespace + eBPF socket quotas.	Budget-exhausting `VBC1` script halts deterministically without impacting sibling sandboxes; Solid LDP pods pass W3C read/write conformance.
WP-05 (P1) Ownership, Family & Cooperative Quorums	`qualia-cooperative-core`, `stewardship.rs`, `identity/agency.rs`	1. Upgrade `derive_lane_key` to Argon2id + HKDF. 2. Wire 1-click `PeerHost::pair` between local `webizen-desktop` (`:8080`) and VPS (`:4242`) with encrypted `.q42j` backup & Shamir CLI recovery. 3. Enable M-of-N Steward Quorum (`SuspendedTransactionQueue`).	Individual, Family (`GuardianshipSemantics`), and M-of-N Cooperative server ownership modes pass audit; domain handle revoke passes `Collapse Detector`.
WP-06 (P1) User-Powered Decentralized CDN (`Q-CDN`)	`services/webtorrent_*.rs`, `net/peer/replication/`, `net/fetch_10d.rs`	1. Replace `mock_network_fetch` in `net/fetch_10d.rs` with live QPR/HTTP range transport. 2. Connect `:4242` WebTorrent seeder & Merkle replication to `webizen-desktop` and browser OPFS caches. 3. Automate HTTP 307 / `Alt-Svc` swarm redirect on origin quota threshold.	Peers serve verified 40 KiB `.q42` SuperBlocks and `.hmc` slices; tampered peer chunks are rejected by SHA-256/CRC-32C before arena admission.

Work Package	Target Crates & Modules	Engineering Deliverables & Integration Tasks	Acceptance & Exit Evidence
WP-07 (P1/P2) Micropayments, Commons Sunset & DLT	`modalities/value_flow.rs`, `services/ilp_dispatcher.rs`, `chronik.rs`	1. Persist `value_flow.rs` obligation pools in `.q42j`. 2. Enforce HTTP `402` / `QPolicy` commercial toll gate (exempting Natural Persons & Humanitarian ICT) with auto `10×` sunset. 3. Stream `ilp_dispatcher.rs` + `chronik.rs` eCash payouts to CDN seeders & prepare Tier 3 UTXO-BFT fork.	Commercial scraper requests settle ILP/eCash micropayments into `.q42` N-Quad receipts; reaching 10× $C_{\text{obligation}}$ permanently flips asset to `EnforcePermissiveCommons`.
WP-08 (P1/P2) Poet Host Admin UI & Turnkey Installer	`crates/poet/`, `webizen-studio`, `qualia-cli`	1. Build the human-centric **Poet Host Admin Workbench** (Domains, Sovereign Relationship Mail, Ontology Compiler, Sandboxes, Hard Cost-Cap Slider & CDN Earnings HUD). 2. Package 1-command Linux VPS installer (`qualia-cli host install`).	Clean Linux VPS provisions full stack in <5 minutes; human operator configures domain, relationship mail, ontology seed, and hard cost cap via UI.

10. References & Authoritative Source Index

- **WebCivics Ontology Repository (>100 MB Seed Target):** https://github.com/WebCivics/ns and live human-rights instrument service https://ns.webcivics.net/
- **QualiaDB & Webizen Core Documentation & Live Portals:** Webizen / QualiaDB Portal (0.0.39), QDNF — Qualia Decentralized Network Fabric, Qualia · Webizen · Poet Release Matrix, Honest Progress 0.0.37, Honest Progress 0.0.33 (VibeScript 0.1), Honest Progress 0.0.26 (Life Domains), and QualiaDB 0.0.40-dev Source Repository
- **Webizen Historical & Architectural Specifications:** DRAFT: Guide For Developers (Church & State Vaults, WebizenID-TLS, Relationship Mail, Permissive Commons), Host Software Requirements (VPS), Webizen Pro VPS & Confidential Computing, and W3C Solid MIT Baseline (web.civics.au)
- **Workspace Architecture & Audit Documents:** Project Definitions (The Webizen Web-Host Server Stack), QualiaDB Capability Reference for Ecosphere Assets (0.0.40 Audit), Ecosphere Assets Specification (Obligation Cost & 10× Commons Sunset), WEBIZEN_REORG_NOTES.md, UI / UX / Human Centric Systems Considerations, Webizen Desktop Application Requirements, and Optimizing WGPU for AI and QualiaDB